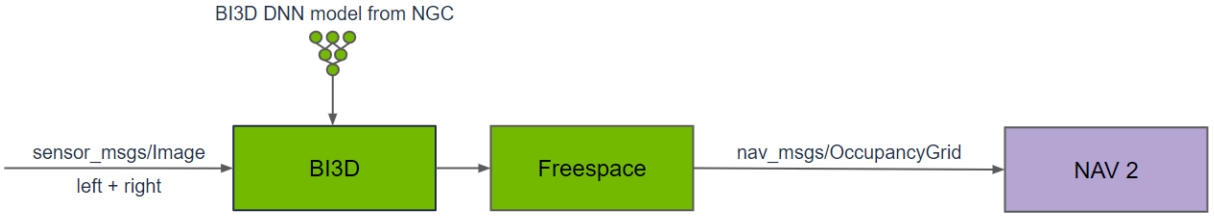
Isaac ROS Free space segmentation

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- Overview
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Isaac ROS Free Space Segmentation official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_freespace_segmentation/index.html

Overview

Free space segmentation processes segmentation masks that contain the pose of the robot relative to the ground, and generates the occupancy grid used for obstacle avoidance by Nav2. Bi3D free space is an additional source of occupancy grid that mobile robots can use.



Quick experience

First start the container with the following command

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open the container terminals with `docker exec -it <container name> bash`. Each terminal has to use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: all the terminals below are executed inside the container:

Terminal A: start the Nano pool=8 ESS. The input image is 1920×1080 and the model input is 960×576:

```
ros2 launch /workspaces/isaac_ros-dev/launch/ess_nano_pool8.launch.py \
  engine_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.1.0_onnx_tr
t10.16/ess.engine \
  ess_plugin_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.1.0_onnx_tr
t10.16/plugins/aarch64/ess_plugins.so \
  input_image_width:=1920 input_image_height:=1080 \
  model_input_width:=960 model_input_height:=576
```

Terminal B: play the input in a loop at 0.25 times the speed:

```
ros2 bag play /workspaces/isaac_ros-dev/isaac_ros_assets/isaac_ros_ess/rosbags/ess_rosbag \
  --loop --rate 0.25 --remap \
  /left/camera_info:=/left/camera_info_rect \
  /right/camera_info:=/right/camera_info_rect
```

Terminal C: run the depth segmentation post-processing:

```
python3 /workspaces/isaac_ros-dev/ess_segmentation.py
```

Terminal D: open rqt image view and select `/ess/depth_segmentation` under Topic:

```
ros2 run rqt_image_view rqt_image_view
```



